MTH1002-A

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

January 2022

Mathematical Methods

Module Leader: Dr Layal Hakim

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No word count

SECTION A

1. (a) Determine the slope of the tangent to the hyperbola $x^2 - y^2 = 9$ at the point $(x, y) = (5, -4)$. (10)

   (b) Find the roots of the complex polynomial

   $$P(z) = z^4 - z$$

   and plot them on an Argand diagram. (10)

   (c) Determine whether the following series converges:

   $$\sum_{n=1}^{\infty} \frac{\cos n}{n^2 - 10}.$$

   (10)

   (d) Find the following limit (if it exists), clearly showing your working out:

   $$\lim_{n\to\infty} \sqrt{3n^2 + n} - \sqrt{3n^2 - n}.$$

   (10)

   (e) Find a vector parallel to the plane $2x + y + z = 1$ and orthogonal to the vector $\hat{i} + \hat{j} + \hat{k}$. (10)

   [**50**]

SECTION B

2. (a) Suppose $z = -1 + i$.

   (i) Find, and simplify, $|z|, 1/z$, arg z and $\ln z$.

   (ii) Write $z = -1 + i$ in polar form and use the result to compute $(-1 + i)^{24}$. (10)

   (b) Consider the three vectors $\vec{a} = (\alpha, 1, 2), \vec{b} = (\beta, 3, 4)$ and $\vec{c} = (\gamma, -1, 2)$. What relationship must α, β and γ satisfy so that $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$? (7)

   (c) The function g is defined by $g(x) = \tan^{-1}(\cos x)$.

   (i) Is g odd, even or neither? Justify your answer.

   (ii) Is g periodic? Justify your answer.

   (iii) What is the domain and range of g? Justify your answer. (8)

   [25]

3. (a) Test the series

$$\sum_{n=1}^{\infty} (-1)^n \frac{n!}{n^n}$$

   for absolute convergence. (7)

   (b) For the function with rule $f(x) = \cos(1/x)$, discuss, with justification

   (i) whether or not the function is defined at $x = 0$;

   (ii) whether or not the function at $x = 0$ is continuous. (8)

   (c) Use Leibniz's rule to differentiate $x^2 \sin x$ five times. (10)

   [25]

4. (a) Locate any maxima or minima for $y = x^{1/3}$. (5)

(b) The function f is defined by

$$f(x) = \sec x = \frac{1}{\cos x}.$$

(i) Sketch the graph of the function.

(ii) Determine the first three terms in the Taylor series of $f(x)$ about $x = \pi/4$. (10)

(c) Consider the planes

$$x + y + z = 3 \quad \text{and} \quad 3x - 2y - z = 4.$$

(i) Show that the planes are orthogonal.

(ii) Find the equation of a plane through the origin which is perpendicular to the line of intersection of these two planes. (10)

[**25**]